



Defending Agentic Federated Learning Systems Against Explanation Poisoning

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Introduction

A **decentralized training paradigm** where raw data never leaves the client devices. Each client trains the model locally and sends only model updates (gradients or weights).

- Data remains local to clients
- Server aggregates updates
- Trust assumption: clients are honest

Core limitation: security is purely numerical, i.e. they detect statistical anomalies, not malicious intent. A carefully crafted adversarial update can still poison the global model and appear statistically normal.

Here, clients are autonomous agents that can perform

- Data selection
- Reward shaping
- Self-reflection

They are also capable of maintaining internal memory & reasoning.

Thus, clients can behave strategically, not just statistically.



Introduction ctd..

Explainability in Federated Learning (FL)

Explainable AI as a Trust Signal:

- Saliency maps
- Feature attributions
- Attention weights
- Concept activation vectors

Used for

- Model auditing
- Debugging
- Regulatory compliance

However, the main assumption here is that a faithful explanation implies a trustworthy model.



Introduction ctd..

Motivation

- **Federated Learning (FL)** is increasingly adopted in privacy-critical and safety-critical domains such as healthcare, autonomous systems, and financial decision-making.
- Modern FL systems are evolving from passive data contributors to **agentic participants** that independently select data, objectives, and training strategies.
- This increased **autonomy** expands the adversarial threat surface, enabling clients to act strategically rather than randomly.
- Agentic clients can launch **stealthy attacks** that preserve model accuracy while manipulating internal logic, trust mechanisms, or explanations.
- Traditional FL defenses, which primarily rely on **aggregation statistics** and **weight-space anomaly** detection, are insufficient to detect such behavior.
- As a result, current FL security mechanisms fail to address intentional, goal-driven adversaries, creating a **critical gap in trust and governance**.



Introduction ctd..

Threat: Explanation Poisoning

Threat

- A new adversarial paradigm in which a model is trained to produce correct predictions and plausible explanations while secretly relying on malicious backdoor logic.
- By explicitly regularizing explanation outputs during training, attackers ensure that post-hoc explainability tools highlight benign features instead of the true trigger.
- As a result, human auditors and automated XAI-based defenses are systematically misled.

Consequences

- The server never observes raw data or internal training behaviour. If explanations can be manipulated, then the system has no reliable signal to distinguish honest agents from adversarial ones.
- This breaks the fundamental trust assumption underlying collaborative learning and allows backdoors to persist across global rounds without detection.



Problem Statement and Objectives

Problem Statement:

Agentic Federated Learning is vulnerable to **Explanation Poisoning**, where adversaries hide backdoors behind fake, plausible explanations (e.g., manipulated heatmaps). Current defenses fail because they audit **weights** (Euclidean distance) rather than **logic**. This allows "Stealth Attacks" to bypass statistical checks, creating models that are accurate but causally corrupt.

Objectives:

- **Formalize Logic-Based Governance:** Shift auditing from Parameter Space (Weights) to Topological Space (Causal Logic).
- **Defeat Explanation Poisoning:** Prove that minimizing Graph Edit Distance creates a mathematical paradox that forces adversaries to disarm their backdoors.
- **Enable Scalable Verification:** Implement Neural Graph Matching to approximate NP-Hard graph validations in real-time.



Literature Review

Base Paper (Published in Elsevier):

P. Riba, A. Fischer, J. Lladós, and A. Fornés, “Learning graph edit distance by graph neural networks,” Pattern Recognition, vol. 120, 2021.

Methodology

- Graph Neural Network encoder
- Siamese / Triplet training framework
- Supervised learning on known GED values
- End-to-end differentiable model



Literature Review

Paper / Work	Authors	Venue	Methodology	Advantages	Disadvantages / Limitations
Scaffolding Attacks on Explainable ML	Liyanage et al.	IEEE (2024)	Introduced <i>Explanation Scaffolding</i> , where model behavior is biased while post-hoc explainers (e.g., SHAP) are manipulated to conceal the bias	Demonstrates that explanation-level manipulation is possible without degrading accuracy	Does not propose any concrete or mathematical defense; relies on post-hoc auditing assumptions
Output Shuffling Attacks on SHAP	Jiawang Liu, Changgen Peng, Weijie Tan, Chenghui Shi	NeurIPS (2024)	Wrapped the target model with a surrogate function that preserves predictions but alters SHAP attributions	Shows that SHAP can be systematically fooled while protected attributes remain influential	Explains the vulnerability but offers no mitigation strategy beyond “stronger auditing”
Federated Backdoor Attacks in Clinical AI	Jun Yuan, Aritra Dasgupta	Nature Medicine (2026)	Empirical study showing that small-scale poisoned updates can implant backdoors in large federated medical models	Highlights extreme real-world risk in healthcare FL systems	Focuses on attack feasibility; explanation integrity and trust mechanisms are not addressed
FedREDefense	Yueqi Xie, Minghong Fang, Neil Zhenqiang Gong	ICML (2024)	Used autoencoder reconstruction error to detect anomalous client updates	Effective against noisy, unstructured, or high-variance attacks	Fails against clean, structured attacks such as Explanation Poisoning



Literature Review ctd..

Paper / Work	Authors	Venue	Methodology	Advantages	Disadvantages / Limitations
ESFL (Efficient & Secure Federated Learning)	Yinbin Miao; Rongpeng Xie; Xinghua Li; Zhiquan Liu; Kim-Kwang Raymond Choo; Robert H. Deng	IEEE TDC (2025)	Combined adaptive local differential privacy with compressive sensing	Improves client privacy and communication efficiency	Privacy noise can mask malicious patterns; no protection against trust or explanation attacks
Triple-Phase Sanitization	Jaehyun Lee, Youngho Cho, Ryungeon Lee, Simon Yuk, Jaepil Youn, Hansol Park and Dongkyoo Shin	MDPI (2026)	Used linguistic statistics (readability, word frequency) to detect poisoned NLP samples	Works well for text-based federated datasets	Domain-specific; does not generalize to vision or medical FL
FedCausal	Dezhi Yang, Xintong He, Jun Wang, Guoxian Yu, Carlotta Domeniconi, Jinglin Zhang	AAAI (2025)	Global DAG optimization under non-IID federated data	Robust causal discovery across heterogeneous clients	Causality used only for performance, not trust or validation
GRAPHEDX	Eeshaan Jain, Indradyumna Roy, Saswat Meher, Soumen Chakrabarti, Abir De	NeurIPS (2024)	Proposed Neural Set Divergence, a neural approximation of Graph Edit Distance (GED)	Makes real-time graph comparison computationally feasible	Not applied to federated learning security or governance



Literature Gap

Despite advances in federated learning security, explainability, and causal modelling, a critical gap exists at their intersection:

1. Existing federated defences verify numerical similarity or output correctness but do not validate internal reasoning.
2. Explanation-based auditing assumes explanations are faithful, an assumption violated by explanation poisoning.
3. Causal reasoning is used for interpretability and fairness but not for adversarial detection.
4. No existing framework enforces consensus over reasoning logic across federated agents.

As a result, federated systems lack a mechanism to distinguish benign autonomy from malicious semantic deviation.



Proposed Solution

A federated learning framework in which each agent submits not only model updates but also a formal proof of its reasoning process in the form of a causal graph. The server validates updates by measuring structural similarity between client reasoning graphs and a global consensus logic

Causal Proof of Reasoning (PoR)

Each client locally extracts a Directed Acyclic Graph (DAG) representing causal dependencies between learned concepts and predictions. During aggregation, the server computes a logic distance metric between the client graph and the global consensus graph. Updates are accepted only if the reasoning structure remains consistent with the federation.

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Algorithm (Client Side)

Input:

Global model weights θ_t
Global causal graph G_{global}
Local dataset D_k

Output:

Structured payload $\{\Delta\theta_k, G_k\}$

Steps

1. Receive broadcast from server containing θ_t and G_{global} .
2. Perception Module (Data Ingestion)
3. Load local dataset D_k .
4. Perform causal filtering to separate invariant causal features from spurious correlations.
5. Initialize local model
6. Set $\theta_k = \theta_t$.
7. Local Training and Optimization
8. Repeat for each epoch:
 7. Compute model predictions.
 8. Calculate task loss (cross-entropy).
 9. Extract temporary causal graph G_{local} .
 10. Compute logic consistency loss between G_{local} and G_{global} .
 11. Update parameters using:
 12. $L_{\text{total}} = L_{\text{task}} + \lambda \times L_{\text{logic}}$.



Algorithm (Client Side) ctd...

13. Cognitive Module (Reasoning Extraction)
14. Analyze gradients and activations.
15. Build adjacency matrix A_k .
16. Construct directed causal graph G_k .
17. Action Module (Payload Construction)
18. Compute weight update:
19. $\Delta\theta_k = \theta_k - \theta_t$.
20. Create payload: $\{\Delta\theta_k, G_k\}$.
21. Upload payload to server.



Algorithm (Server Side)

Input:

Client payloads $\{\Delta\theta_k, G_k\}$
Global model θ_t
Global consensus graph G_{global}

Output:

Updated global model θ_{t+1}
Updated global causal graph $G_{\text{global_new}}$

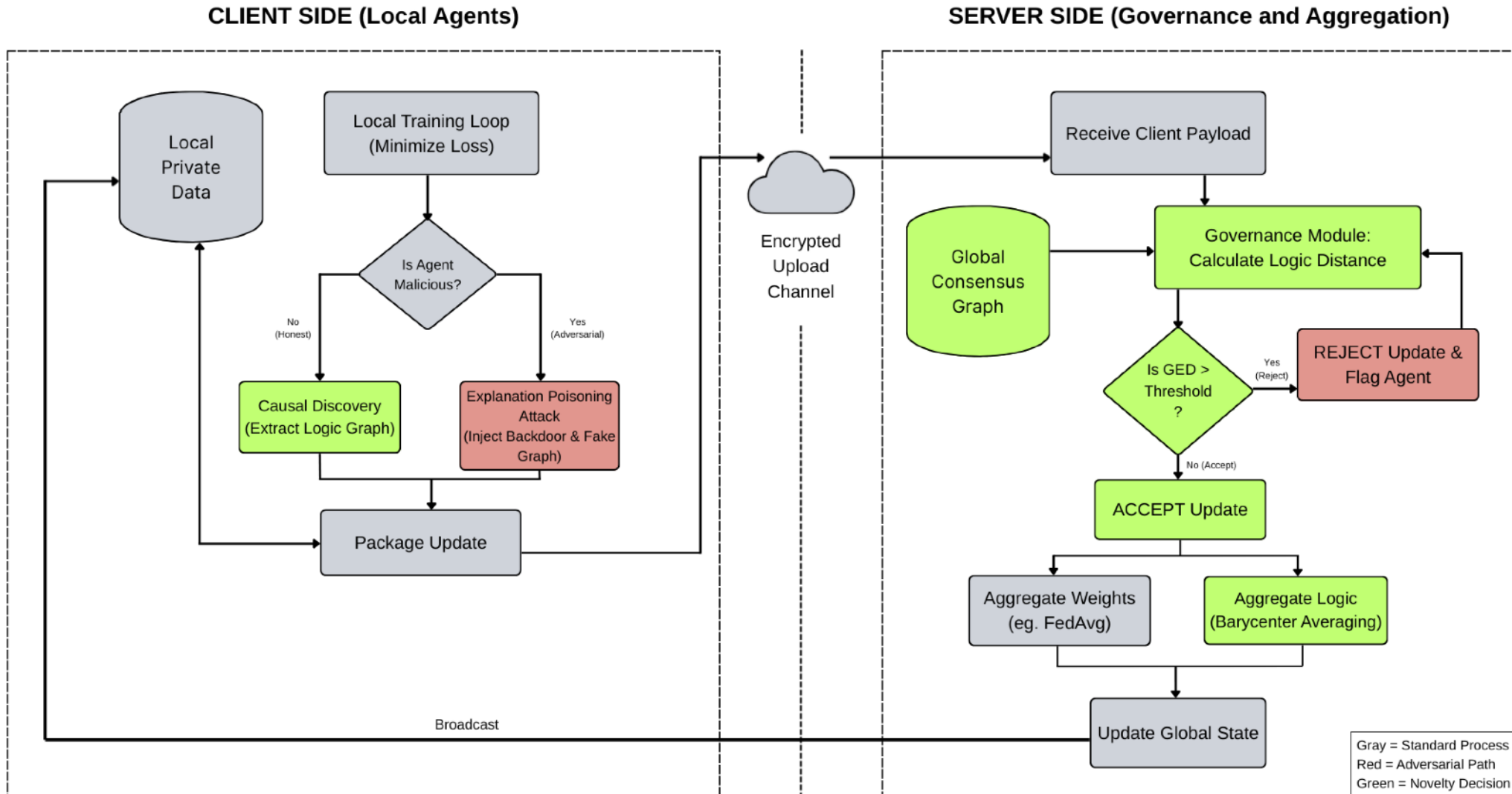
Steps

- 1) Broadcast Phase
- 2) Send θ_t and G_{global} to all clients.

- 3) Collect Updates
- 4) Receive payloads from all participating clients.
- 5) Logic Validation using Neural GED
- 6) Initialize Accepted set as empty.
- 7) For each client k :
- 8) Compute logic score using Siamese GNN:
- 9) $\text{score}_k = ||\text{GNN}(G_k) - \text{GNN}(G_{\text{global}})||^2$
- 10) If $\text{score}_k \leq \tau$, accept update.
- 11) Otherwise reject update as logically anomalous.
- 12) Dual Aggregation
- 13) Weight Aggregation
- 14) Compute new global weights using accepted clients:
- 15) $\theta_{t+1} = \sum (n_k / N) \times \theta_k$.
- 16) Logic Aggregation
- 17) For each edge in accepted graphs:
- 18) Keep edge if it appears in more than 50% of graphs.
- 19) Form updated consensus graph $G_{\text{global_new}}$.
- 20) Update Global State
- 21) Set $\theta_t = \theta_{t+1}$.
- 22) Set $G_{\text{global}} = G_{\text{global_new}}$.
- 23) Begin next federated learning round.



Block Diagram





Novelty

This is the first federated learning framework that enforces structural reasoning consistency rather than statistical similarity. The approach is explanation-agnostic, model-agnostic, and directly targets the root cause of explanation poisoning attacks. It transforms trust verification from a heuristic process into a formal governance mechanism.

The Graph Edit Distance (GED) between a client graph G_c and the global consensus graph G_g is defined as:

$$GED(G_c, G_g) = \min_{(op_1, \dots, op_k) \in \Gamma(G_c, G_g)} \sum_{i=1}^k \text{Cost}(op_i),$$

where $\Gamma(G_c, G_g)$ is the set of valid edit-operation sequences transforming G_c into G_g .

Allowed Operations:

- **Vertex substitution:** $v \rightarrow u$ (changing a feature node),
- **Edge insertion/deletion:** e^+, e^- (adding or removing a causal link).

This is an NP - Hard problem - So we fix it by using Siamese GNN to calculate the distance between the causal graph generated by each node with the global consensus in polynomial time

Disclaimer: This is a screenshot from our report and not from a paper that exists



Dataset Selected

1. CelebA (Visual Logic Validation)

- What: 202k+ Face Images with 40 dense binary attributes (e.g., Smiling, Eyeglasses).
- Why: Perfect for testing "Causal Consistency." Attributes follow strict logical rules (e.g., Mouth Open → Smiling), allowing us to simulate and detect "Logic Attacks" where triggers override natural features.

2. FLamby (Healthcare Application)

- What: Realistic suite of 7 medical datasets (Histopathology, MRI, ISIC-2019 Dermoscopy) representing distinct hospital silos.
- Why: The gold standard for Cross-Silo FL. It validates our defense in high-stakes, non-IID environments where "Right Answer, Wrong Reason" (e.g., artifact learning) is fatal, bridging the gap between theory and clinical deployment.



References

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Signatures

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THANK YOU